§2 Formal definition of an ACS — part 5 of 5

Reference: subsec:three_primitive_functions to subsec:assumption_list_sec2

Source: paper/section2_definition.tex, lines 3132–4651 (§2.9–§2.13), at commit e6e12cd. The Japanese manuscript is authoritative. Parts 1–4 are 02_definition_1.md to 02_definition_4.md.

2.9 Basic and base functions, CR-2, and the standard map forms

§2.9 defines the **basic functions** of a realization of the 7-tuple structure, with respect to the primitive functions defined in §2.3 (subsec:primitive_functions), and gives the constitutive requirement **CR-2**, which requires them. It then states the standard map forms P_{sens} (sensing), P_{disc} (discrimination), P_{val} (evaluation) and P_{act} (action) used to analyze the primitive functions in the paper's formal system, together with their non-degeneracy conditions and their operational dependencies.

CR-2 is one of the two constitutive requirements forward-referenced by def:acs; the other, CR-1, was given in §2.6. Both are conditions for placing a 7-tuple structure in the class of ACSs in this paper, but their objects differ. CR-1 constrains each element of the required environment space E to be necessary for persistence. CR-2 requires basic functions corresponding to each class of primitive function, their functional non-degeneracy, and the base functions that support their activity. The paper does not state either as a general consequence of the other.

New symbols: the space $\mathcal{J}$ of sensed information, the space $\mathcal{R}_{\text{int}}$ of internal representations, and the set $\mathcal{V}$ of evaluation values. The subscript int stands for *internal*.

Definition (def:basic_functions): basic functions. The functions that a realization of the 7-tuple structure Σ of def:acs has with respect to each class of the primitive functions (def:primitive_functions) are called the **basic functions** of that realization (for an ACS individual Σ_i, the basic functions of Σ_i). Basic functions are a description of functional responsibility. They do not mean a particular physical part, computational module, standard map or one-to-one implementation split. It is not excluded that several basic functions jointly bear the responsibility of one class, or that one basic function takes part in the responsibilities of several classes. The object of the definition is "a realization of the 7-tuple structure" rather than "an ACS individual" because CR-2, below, uses this definition as a condition for being an ACS.

Constitutive requirement CR-2

Reference: req:CR2

As a constitutive requirement for being an ACS, a realization of the 7-tuple Σ is required to satisfy (CR-2a)–(CR-2c).

- **(CR-2a) Possession of basic functions.** For each of the four classes of primitive function (sensing, discrimination, evaluation, action), the realization possesses a basic function (def:basic_functions).
- **(CR-2b) Functional non-degeneracy.** The basic functions of (CR-2a) are non-degenerate for each class, in the following sense.
  1. *Sensing:* with the external environment state fixed, there is at least one case in which different internal states yield different information; and with the internal state fixed, there is at least one case in which different external environment states yield different information.
  2. *Discrimination:* in at least one internal state, there is a case in which a difference of external environment state leads, through the information obtained, to different internal representations.
  3. *Evaluation:* there is at least one case in which different input objects of evaluation are treated as non-equivalent. Further, when the feasible set K (def:K) satisfies $K \neq \emptyset$ and $X \setminus K \neq \emptyset$, there is at least one case in which, through sensing and discrimination, the evaluation differs between a state belonging to K and one not belonging to it (**feasibility sensitivity**). In the standard map form the comparison is made under the same internal-model state and external environment state.

4. *Action:* there is at least one case in which the generated response differs according to a difference in at least one of the available internal representation, the evaluation, or the current state. It is not required that the response depend on all of these differences.

- **(CR-2c) Possession of base functions.** The realization possesses the base functions (`def:base_functions`) that support the activity of the basic functions of (CR-2a).

Possession in (CR-2a) and (CR-2c) means that the function is present as part of the realization's constitution; it is not required that the function be operating at each time. (CR-2b) likewise is a condition that the possessed functions are not degenerate, not a condition on operation at each time. Cessation of operation at a time is treated by the judgment of functional termination (`def:functional_termination`).

(CR-2b) is stated at the level of classification A (functional conditions) and is not given as a property of a particular map. The non-degeneracy conditions of `def:three_primitives` express each item of (CR-2b) as properties of maps when the paper describes a system using the standard map forms. Matching the basic functions of a realization with the standard map forms is a choice in describing the target system, and Phase 1 gives no general criterion for it.

CR-2 introduces no independent component added to the 7-tuple; it is a condition for the 7-tuple Σ to qualify as an ACS. `def:three_primitives` gives the standard map forms for analyzing the functional responsibilities of the primitive functions. CR-2 does not require a realization to have those standard maps themselves as parts; functionally equivalent realizations meeting the same responsibilities are allowed. $P_{\text{sens}}, P_{\text{disc}}, P_{\text{val}}, P_{\text{act}}$ are neither components of the 7-tuple $\Sigma = (X, U, E, f, \Pi, M, \text{Update})$ nor uniquely derived from it. A system lacking, in its constitution, a basic function for some class of primitive function does not satisfy the paper's definition of an ACS.

CR-2 is a condition identifying Σ_i as an ACS individual, not a diagnosis at each time (`rem:post_collapse_record`). Therefore, if an identified Σ_i later loses, through damage or otherwise, the basic functions of one or more classes wholesale and without recovery, that alone does not cancel its identification as an ACS individual. The paper does not re-judge CR-2 at such a time to declare Σ_i not an ACS, nor retroactively cancel the earlier identification. The functional state of Σ_i is then described by whether functional termination holds (`def:functional_termination`) and by the descriptive state bands of §7. Wholesale loss of one or more classes is, by itself, neither functional termination nor collapse (`rem:termination_hierarchy`).

CR-2 is not an assumption limiting the range in which later propositions hold; it is a constitutive requirement defining the object "ACS". Hence, like CR-1 and the convention TSC, it is not included in the ordinary list of assumptions but treated as a foundational condition supporting the definition of an ACS.

Constitutive necessity of the primitive functions (`rem:three_primitives_necessity`, classification A). For an ACS that aims at maintaining long-term feasibility in the environments described in §1, the following functional responsibilities are constitutively required for each class of primitive function: obtaining information from its own internal state and the external environment state (sensing); distinguishing the information obtained and forming internal representations (discrimination); treating internal representations, its current state, candidate responses and other objects of evaluation non-equivalently against an internal criterion concerning its own state (evaluation); and generating a response according to at least one of the available internal representation, the evaluation, or the current state (action). For evaluation and action, using all of the listed objects or information is not required (CR-2b; `def:three_primitives`). Evaluation is set up as a class because sensing, discrimination and action alone contain no responsibility for tying response generation to a criterion concerning the system's own state, so the responsibility embodied in "aiming at" long-term feasibility would be missing. The evaluation class is a concept at the level of function; it is consistent with the evaluation-related ones among the principle-related structural functions of §3, but it does not express the truth or falsity of the time-indexed satisfaction predicates of the three principles (`rem:primitive_vs_principles`). The internal criterion of evaluation is not the persistence evaluation function Π, and the paper does not require an ACS to hold or compute Π internally.

The function of retaining internal representations across times is distinguished from the

classes of primitive function and is treated under the internal model structure (§2.10) and in §4. Insofar as it supports the activity of basic functions, it can be included in the base functions (`def:base_functions`); in that case it is an object of condition (FT-1)(b) of functional termination (`def:functional_termination`). Internal-model update is not added as a class; where it bears the responsibility of some class in a target system, it can be treated as a basic function for that class. These functional requirements are not required to be implemented as independent physical parts, computational modules or explicit maps, and functionally equivalent realizations are allowed. The remark does not claim that the necessity of these responsibilities has been proved for all mathematical environments; its scope follows the stipulation of §1. `def:three_primitives`, below, is the standard map form adopted for analyzing, comparing and describing these functional requirements in the paper's formal system. What CR-2 requires is possession of basic functions for each class, their functional non-degeneracy (CR-2b) and possession of base functions, not the implementation of a particular map decomposition; the non-degeneracy conditions of that definition express (CR-2b) on the standard map form.

Definition (`def:three_primitives`): standard map forms of the primitive functions (classification B). Let $\mathcal{I}, \mathcal{R}_{\mathrm{int}}, \mathcal{V}$ be nonempty sets serving as the types of sensed information, internal representations and evaluation values. The following total, single-valued maps are fixed:

$$P_{\mathrm{sens}} : X \times E_{\mathrm{ext}} \to \mathcal{I},$$
$$P_{\mathrm{disc}} : \mathcal{M} \times \mathcal{I} \to \mathcal{R}_{\mathrm{int}},$$
$$P_{\mathrm{val}} : \mathcal{M} \times \mathcal{R}_{\mathrm{int}} \to \mathcal{V},$$
$$P_{\mathrm{act}} : \mathcal{M} \times \mathcal{R}_{\mathrm{int}} \times \mathcal{V} \to U.$$

They are standard maps describing the primitive functions; they are neither the functional classes themselves nor additional components of the 7-tuple, and are not claimed to be uniquely derived from the 7-tuple alone. The definition imposes no topology or order on $\mathcal{I}, \mathcal{R}_{\mathrm{int}}, \mathcal{V}$ and requires no continuity of the maps. An evaluation value may be a bundle containing per-candidate evaluation information. No direct access of the internal mechanism to the true state or environment is required, and the same-time inputs are connected as

$$I_t = P_{\mathrm{sens}}(x_t, e_t),$$
$$r_t = P_{\mathrm{disc}}(M_t, I_t),$$
$$v_t = P_{\mathrm{val}}(M_t, r_t),$$
$$u_t = P_{\mathrm{act}}(M_t, r_t, v_t).$$

An internal representation may include components retaining available information; substantive dependence on every argument is not required.

In a description using the standard maps, the non-degeneracy of CR-2b is expressed by the following conditions.

1. *Sensitivity of sensing to internal-state differences:* $\exists e \in E_{\mathrm{ext}},\ x, x' \in X$: $P_{\mathrm{sens}}(x, e) \neq P_{\mathrm{sens}}(x', e)$.
2. *Sensitivity of sensing to external-environment differences:* $\exists x \in X,\ e, e' \in E_{\mathrm{ext}}$: $P_{\mathrm{sens}}(x, e) \neq P_{\mathrm{sens}}(x, e')$.
3. *Discrimination:* for some $M \in \mathcal{M},\ x \in X,\ e, e' \in E_{\mathrm{ext}}$: $P_{\mathrm{disc}}(M, P_{\mathrm{sens}}(x, e)) \neq P_{\mathrm{disc}}(M, P_{\mathrm{sens}}(x, e'))$.
4. *Evaluation:* P_{val} is non-constant on its domain. Further, when $\emptyset \neq K \neq X$, for some $M \in \mathcal{M},\ e \in E_{\mathrm{ext}},\ x \in K,\ x' \in X \setminus K$: $P_{\mathrm{val}}(M, P_{\mathrm{disc}}(M, P_{\mathrm{sens}}(x, e))) \neq P_{\mathrm{val}}(M, P_{\mathrm{disc}}(M, P_{\mathrm{sens}}(x', e)))$.
5. *Action:* P_{act} is non-constant on its domain.

The sensing conditions imply $|X| \geq 2$, the discrimination condition $|\mathcal{R}_{\mathrm{int}}| \geq 2$, the evaluation condition $|\mathcal{V}| \geq 2$, and the action condition $|U| \geq 2$. Even if $\mathcal{M}$ is a single point, non-degenerate operation through differences in information is allowed. Feasibility sensitivity is an existence condition comparing under the same M, e; it does not require discrimination of all state differences, accuracy of evaluation, monotone correspondence with Π, or selection of feasible actions. For $K = \emptyset$ or $K = X$, no additional feasibility-sensitivity condition is imposed; for $\emptyset \neq K \neq X$ the additional condition implies non-constancy of evaluation. The paper does not state, as a general consequence of non-constancy of evaluation and action alone, non-constancy on the input images actually arising at run time. For selective waiting $u_\emptyset$ and the

distinction of directed internal and external actions, see rem:action_space_structure; the autonomous functional process f_{auto} is not an object of selection.

Proposition (prop:feasibility_sensitivity_scope): the range over which feasibility sensitivity imposes a requirement. Under ass:env_viable_nondegeneracy, $K \neq \emptyset$ and $X \setminus K \neq \emptyset$. Hence, under this assumption, the feasibility sensitivity of evaluation in def:three_primitives is not the case of "requiring nothing" but a condition that imposes a requirement. *Proof outline.* By the assumption there are x and $e \neq e'$ with $A(x,e) \neq A(x,e')$; without loss of generality take $u \in A(x,e) \setminus A(x,e')$ (the same inference as in the proof of prop:nondegeneracy_implication). By def:at, $f(x,u,e) \in K$ and $f(x,u,e') \notin K$; the former gives $K \neq \emptyset$, and since f maps into X, $f(x,u,e') \in X \setminus K$. CR-2 is a condition fixing the class of ACSs independently of the assumptions of §2; the proposition does not change CR-2 and says only that, within the range using that assumption, feasibility sensitivity is required of evaluation.

Sensing non-degeneracy versus the other levels (rem:sensing_nondegeneracy_relation). The non-degeneracy of sensing with respect to external-environment input in def:three_primitives is the third level of non-degeneracy with respect to external-environment input (rem:nondegeneracy_levels); the first concerns state transition (ass:env_input_nondegeneracy) and the second the feasible action set (ass:env_viable_nondegeneracy). Unless a connecting condition is separately placed, the paper assumes no mutual implication between the sensing level and the other two. This is because no connecting condition is placed between the standard sensing map P_{sens} and the state-transition map f; it is not a claim that the two have been proved independent. Sensing non-degeneracy is also a condition separate from CR-1 of the required environment space E: that a difference of external environment state is sensed, and that the difference concerns satisfaction of a required environmental factor, are distinguished in the paper.

Operational dependence of the standard maps (rem:P_dependency). The composition in def:three_primitives is a same-time connection convention for information generation; it does not derive that particular map decomposition from the possession of functions in CR-2. Feasibility sensitivity requires the transmission of differences through this connection. P_{act} need not use all components of r_t, ν_t. In a standard form including local protective responses, local information is made available as a component of r_t, so a response not using the result of detailed evaluation can be represented. The composition convention requires no physical sequential execution or lower bound on delay.

Operational dependence versus formal existence (rem:functional_vs_existential). rem:P_dependency states functional operational dependence, not dependence of formal existence. The map $P_{\text{disc}} : \mathcal{M} \times \mathcal{J} \to \mathcal{R}_{\text{int}}$ is definable as a type independently of the existence of P_{sens}; likewise P_{val} and P_{act}. What the remark claims is the dependence order by which, inside an ACS, each function operates on meaningful input, not an order of the mathematical existence of the maps.

Information dependence versus physical time order (rem:primitive_no_temporal_order). The same-time composition order specifies which output each input comes from; it does not fix the implementation of each map to a physical sequential module. M_t may retain information from earlier times, but the same-time r_t, ν_t are not required to be already contained in M_t. The effect of the effective action can be reflected in later sensing through $x_{t+1} = f(x_t, u_t, e_t)$. A non-degenerate dependence of f on its action argument is not required in general (ass:f_decomposition).

2.10 The internal model structure M

Reference: subsec:internal_model_M

§2.10 defines the internal model structure M with which an ACS processes the environment, its own state and action possibilities. The paper treats M as decomposed into three mechanisms — information integration, adaptive evaluation and response action — and calls this three-component structure the **Triadic Information-Processing Architecture (TIPA)**. TIPA is a general term for the mechanism composition and interconnection of M_1, M_2, M_3 and is not a new independent component of the 7-tuple. The three mechanisms of TIPA are distin-

guished both from the abstract functional classes called primitive functions and from the standard map forms of `def:three_primitives`; no one-to-one correspondence is placed between the three mechanisms and the classes of primitive function. The operating principle and interconnection of each mechanism are in §4; this subsection gives the type and basic properties of M as one component of the 7-tuple. It also fixes the type of the internal environment estimate, delegated from §2.8, and the treatment of encoding of internal representations, delegated from §2.9.

New symbols: the space $\mathcal{M}$ of internal-model states, its product components $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$, the product distance $d_{\mathcal{M}}$, the TIPA component-wise change $\delta_{M,c,t}$, the TIPA component-wise change vector $\boldsymbol{\delta}_{M,t}$, the maximum aggregated change δ_t, and the internal environment estimate $\hat{e}_t$.

Type and state space of the internal model

Assumption (`ass:M_compact`): compactness of $\mathcal{M}$ and the scale convention for component distances. For each $c = 1, 2, 3$, nonempty compact metric spaces $(\mathcal{M}_c^{\mathrm{L}}, d_c^{\mathrm{L}})$ and $(\mathcal{M}_c^{\mathrm{W}}, d_c^{\mathrm{W}})$ are given, and

$$\mathcal{M}_c = \mathcal{M}_c^{\mathrm{L}} \times \mathcal{M}_c^{\mathrm{W}},$$
$$d_c((l, w), (l', w')) = \max\{d_c^{\mathrm{L}}(l, l'),\ d_c^{\mathrm{W}}(w, w')\}.$$

The superscripts L, W are identifying labels for the learning component and the working component, not powers. Each factor carries its metric topology and the product the product topology. If a factor has more than one point, its metric is normalized to diameter 1; if a single point, its distance is 0. A finite product is compact, and the max distance above gives the product topology; if either factor has more than one point, the diameter of d_c is 1, and if both are single points it is 0. The decomposition is a specification of state carriers; it does not claim independence of the updates of the learning and working components, separation of time scales, or isometry with any existing metric.

Each component $\mathcal{M}_c$ $(c = 1, 2, 3)$ of the internal-model state space is thus a nonempty compact metric space $(\mathcal{M}_c, d_c)$; nonemptiness is needed so that an internal-model state $M_{c,t} \in \mathcal{M}_c$ exists at each time. To avoid an arbitrary choice of distance scale among the three components affecting the maximum-aggregated distance, each component distance d_c is adopted as normalized so that $\mathrm{diam}_{d_c}(\mathcal{M}_c) = 1$ if $|\mathcal{M}_c| \geq 2$, and $d_c \equiv 0$ if $\mathcal{M}_c$ is a single point. This scale convention does not assume that the three mechanisms have equal functional importance or implementation cost; it is a metric convention making component distances comparable on a common relative scale. *Introduction status: S. Strength: strong.* From the compactness of each component follows the compactness of the finite product $\mathcal{M} = \mathcal{M}_1 \times \mathcal{M}_2 \times \mathcal{M}_3$, and the finiteness of each diameter makes the normalization possible. The product structure does not deny the information dependence $M_1 \to M_2 \to M_3$. Relaxation such as local compactness is a Phase 2 matter.

Definition (`def:model_learning_working`): learning and working components of the internal-model state. Put $\mathcal{M}^{\mathrm{L}} := \prod_{c=1}^{3} \mathcal{M}_c^{\mathrm{L}}$ and $\mathcal{M}^{\mathrm{W}} := \prod_{c=1}^{3} \mathcal{M}_c^{\mathrm{W}}$, and let $M_t^{\mathrm{L}} = (M_{c,t}^{\mathrm{L}})_{c=1}^{3}$ and $M_t^{\mathrm{W}} = (M_{c,t}^{\mathrm{W}})_{c=1}^{3}$ be the coordinate projections of M_t. The learning component carries persistent content, learning and forgetting; the working component carries short-term history carried over to later times, records of the previous time's evaluation and action, and the like. Which record is retained in which mechanism component is specified according to the dependence constraints of the update maps. That a readout from $\mathcal{M}^{\mathrm{W}}$ agrees with the actual past value, or that arbitrary histories can be encoded, does not follow from this product structure. $\mathcal{M}^{\mathrm{L}}$ and $\mathcal{M}^{\mathrm{W}}$ are an organization of the coordinates of $\mathcal{M}$, not states added to the 7-tuple.

Definition (`def:internal_model`): internal model structure. The internal model structure is $M = (M_1, M_2, M_3)$. M_1 is the **information-integration mechanism**, responsible for integrating sensed information, boundary management and maintaining temporal continuity. M_2 is the **adaptive-evaluation mechanism**, responsible for generating mechanism-level proxy evaluations related to the three principles, Priority warnings and the like. M_3 is the **response-action mechanism**, responsible for organizing information needed for action decision, composing candidates, using evaluation results and deciding the TIPA action command.

M_1, M_2, M_3 can operate functionally in parallel (ass:functional_concurrency, §4). This functional concurrency is compatible with the generation dependence $M_1 \to M_2 \to M_3$ in which the needed upstream output becomes available as input to the downstream mechanism. The order $M_1 \to M_2 \to M_3$ in the update formulas of def:update_operator expresses information dependence and does not mean that the three mechanisms operate only sequentially in real time. This subsection does not state that the three mechanisms interact at the same time; whether information is exchanged between mechanisms at the same time, and the time convention for it, are treated in §4. Detailed definitions, operating principles and interconnection of each component are given in §4.

Definition (def:model_parameter_space): internal-model state space. Under ass:M_compact, the internal-model state space corresponding to M is $\mathcal{M} = \mathcal{M}_1 \times \mathcal{M}_2 \times \mathcal{M}_3$, each $\mathcal{M}_c$ being the nonempty compact metric space $(\mathcal{M}_c, d_c)$ of that assumption, with $\mathrm{diam}_{d_c}(\mathcal{M}_c) = 1$ if $|\mathcal{M}_c| \geq 2$ and $d_c \equiv 0$ if single-point. This normalization aligns the multiplicative scale of the component distances to a common relative range and does not mean the three mechanisms have equal functional importance, implementation cost or physical amount of change. For $m = (m_1, m_2, m_3)$ and $m' = (m'_1, m'_2, m'_3)$ in $\mathcal{M}$ the **product distance** is

$$d_{\mathcal{M}}(m, m') := \max\{d_1(m_1, m'_1), d_2(m_2, m'_2), d_3(m_3, m'_3)\}.$$

The internal-model state at time t is $M_t = (M_{1,t}, M_{2,t}, M_{3,t}) \in \mathcal{M}$ with $M_{c,t} \in \mathcal{M}_c$. By ass:M_compact and the property of finite products, $\mathcal{M}$ is compact. The product distance $d_{\mathcal{M}}$ is a metric describing the metric-space structure and compactness of $\mathcal{M}$; it is not interpreted as a quantity fully expressing the content of change of the TIPA mechanisms. The dynamic change of each mechanism is described separately by the TIPA component-wise change vector $\boldsymbol{\delta}_{M,t}$ of def:tipa_component_change.

M kept as an independent component (rem:M_independence). The paper does not assume that the internal model structure M can be uniquely recovered from the state-transition map f. $M = (M_1, M_2, M_3)$ gives a mechanistic interpretation to the three-component decomposition of $\mathcal{M}$. What $P_{\mathrm{act}} : \mathcal{M} \times \mathcal{R}_{\mathrm{int}} \times \mathcal{V} \to U$ and Update : $\mathcal{M} \times X \times E_{\mathrm{ext}} \to \mathcal{M}$ act on directly is not the mechanism structure M itself but an element M_t of its state space $\mathcal{M}$. Under a representation policy that explicitly separates the structural roles of f and M, the paper keeps M as an independent component of the 7-tuple. This is not a general theorem that M cannot be derived from f. Formal study of uniqueness or non-uniqueness of the M corresponding to the same f is left to §4 onward, once the concrete maps of TIPA are fixed, and to Phase 2.

Internal environment estimate

Definition (def:internal_env_estimate): internal environment estimate (classification B). The estimate of the external environment state formed by the internal model at time t is written $\hat{e}_t \in E_{\mathrm{ext}}$ and called the **internal environment estimate**; the superscript hat is an identifier for an estimate. $\hat{e}_t$ takes values in the same space E_{ext} as the realized external environment state e_t, but the paper does not assume that the two agree. $\hat{e}_t$ is not a component of the 7-tuple; it is a derived quantity generated by the processing of M_1 on the basis of available mechanism inputs including the internal-model state M_t. The paper does not require $\hat{e}_t$ to be determined by M_t alone. The concrete map generating $\hat{e}_t$ is placed in M_1, and its form is given in §4.

Gap between the realized state and the estimate (rem:estimate_gap). def:internal_env_estimate requires agreement, closeness or any error bound between e_t and $\hat{e}_t$ in no sense. The distinction keeps the system-level description based on the realized external environment state apart from the mechanism-level description based on the environment estimate generated by the internal model. Since Phase 1 introduces no distance, pseudo-distance or estimation-error evaluation map on E_{ext}, the gap is not quantified; quantification of estimation error and evaluation of its influence on maintaining feasibility are left to §4 and Phase 2 (L6 of §9). The state transition $x_{t+1} = f(x_t, u_t, e_t)$ (def:state_transition) contains the realized state e_t, not $\hat{e}_t$.

Retention of internal representations in the internal-model state (`rem:M_encodes_representation`). The same-time internal representation and evaluation value are intermediate outputs of `def:three_primitives`. When part of them is reused at later times, the retained content is read out from the learning or working component of `def:model_learning_working`. The readout rule is fixed for each description of a target, and no recovery of past true values, retention of arbitrary history, or injective, surjective or continuous encoding is required. Writing to each component follows the type of `def:update_operator`. That the comprehensive type has enough arguments does not by itself imply that arbitrary records can be realized by the restricted component-wise updates.

Mechanism components and TIPA integration

Definition (`def:tipa_component_change`): TIPA component-wise change and maximum-aggregated distance (classification C). For M_t and M_{t+1} in $\mathcal{M}$, the one-step change of each TIPA mechanism is

$$\delta_{M,c,t} := d_c(M_{c,t+1}, M_{c,t}), \quad c = 1, 2, 3,$$

and the **TIPA component-wise change vector** is $\boldsymbol{\delta}_{M,t} := (\delta_{M,1,t}, \delta_{M,2,t}, \delta_{M,3,t}) \in \mathbb{R}^3_{\geq 0}$. It is the primary dynamic descriptive quantity describing the changes of M_1, M_2, M_3 without aggregating them with one another. It is introduced for analysis and the ACS is not required to hold it internally. By the scale convention of `ass:M_compact`, $0 \leq \delta_{M,c,t} \leq 1$, so $\boldsymbol{\delta}_{M,t}$ describes, component by component, how much each mechanism changed relative to its whole state space; the normalization does not mean the mechanisms have equal functional importance or implementation cost. The max-aggregated value by the product distance on $\mathcal{M}$ is

$$\delta_t := d_{\mathcal{M}}(M_{t+1}, M_t) = \max\{\delta_{M,1,t}, \delta_{M,2,t}, \delta_{M,3,t}\} = \|\boldsymbol{\delta}_{M,t}\|_\infty.$$

So δ_t is a secondary aggregate expressing the largest one-step change among the three mechanisms, not the whole component information of the change. In particular $\delta_t = 0 \iff \boldsymbol{\delta}_{M,t} = (0, 0, 0)$, but different component-wise vectors can give the same δ_t. Hence in Phase 1, when analyzing the dynamic change of TIPA, $\boldsymbol{\delta}_{M,t}$ is used as the primary description, and $d_{\mathcal{M}}$ and δ_t are used when the distance of the product space, compactness, and aggregation of the maximum component change are needed (within present framework).

Observation (`obs:tipa_integration`): policy of functional separation of $M_1/M_2/M_3$. The paper describes the three mechanisms as constituents bearing different functional roles. First, the separation of M_1 and M_2 is kept to distinguish the mechanism that integrates information and estimates state from the one that evaluates feasibility and warnings on that information; it underlies the evaluation structure including the Priority 3 warning defined in §4. Second, the separation of M_2 and M_3 is kept to describe the mechanism that evaluates and the one that generates responses into the action space U as different functional roles; the domains and codomains of their concrete maps and their interconnection are in §4. The observation does not claim that integrating two mechanisms is mathematically impossible, or that any integration necessarily causes functional degradation. Keeping the three separate is a descriptive composition policy of TIPA in Phase 1. Reformulations permitting mechanism integration and their functional equivalence are left to Phase 1.α onward (within present framework).

Relation between M and Update **(`rem:M_update_relation`).** The internal model structure M and the rule Update that updates its state (§2.11) are kept in the 7-tuple as independent components with different structural roles. The internal-model state at time t is $M_t \in \mathcal{M}$, and P_{act} takes M_t and the same-time r_t and v_t as input to generate an action. Through its state space $\mathcal{M}$ and the state M_t, M is therefore positioned as the mechanistic basis supporting the ACS's action generation. The basic formal relation between M and Update is fixed in §2.11; their correspondence with the standard mechanism forms of TIPA is in §4, and matters concerning change between times are in §5.

2.11 The internal-model update operator Update

Reference: subsec:update_operator

§2.11 defines the operator Update that updates the internal-model state component by component, and introduces the time average of the internal-model change that actually occurred through updating. This is a descriptive quantity of realized change and not a quantity expressing cognitive processing capability (§2.12). The one-step change quantities $\delta_{M,c,t}$, $\boldsymbol{\delta}_{M,t}$ and δ_t were defined in §2.10 (def:tipa_component_change) and are not redefined here; they are referred to as the one-step change produced by Update.

New symbols: the maximum-component time-average update rate $\lambda_{T,t}$, and the component-wise time-average update rates $\lambda_{M,c,T,t}$ and $\boldsymbol{\lambda}_{M,T,t}$. $T \in \mathbb{N}$, $T \geq 1$ is the length of the finite time window used for evaluation. When the window starts at time 0, these are abbreviated λ_T, $\lambda_{M,c,T}$, $\boldsymbol{\lambda}_{M,T}$.

Definition of the update operator

Definition (def:update_operator): structure of the Update operator.

(1) *Comprehensive map* (the details of the type declaration are in rem:update_type_two_layers):

$$\mathrm{Update} : \mathcal{M} \times X \times E_{\mathrm{ext}} \longrightarrow \mathcal{M}.$$

In this comprehensive type the input of Update is limited to (M_t, x_t, e_t). The same-time r_t, v_t, u_t are determined from (M_t, x_t, e_t) under the fixed standard maps, so no independent argument need be added to the comprehensive type to refer to them. Whether each component update can refer to that information is constrained by the detailed types below. An observation newly acquired after the transition is treated as input of the next time and is not anticipated.

(2) *Component-wise updates* (under the product decomposition of ass:M_compact):

$$\begin{gathered}\mathrm{Update}_1 : \mathcal{M}_1 \times X \times E_{\mathrm{ext}} \to \mathcal{M}_1,\\ \mathrm{Update}_2 : \mathcal{M}_2 \times X \times E_{\mathrm{ext}} \times \mathcal{M}_1 \to \mathcal{M}_2,\\ \mathrm{Update}_3 : \mathcal{M}_3 \times X \times E_{\mathrm{ext}} \times \mathcal{M}_1 \times \mathcal{M}_2 \to \mathcal{M}_3,\end{gathered}$$

$$\begin{aligned}M_{1,t+1} &= \mathrm{Update}_1(M_{1,t}, x_t, e_t),\\ M_{2,t+1} &= \mathrm{Update}_2(M_{2,t}, x_t, e_t, M_{1,t+1}),\\ M_{3,t+1} &= \mathrm{Update}_3(M_{3,t}, x_t, e_t, M_{1,t+1}, M_{2,t+1}).\end{aligned}$$

That Update_2 depends on $M_{1,t+1}$ and Update_3 on $M_{1,t+1}$ and $M_{2,t+1}$ reflects the information dependence $M_1 \to M_2 \to M_3$ of TIPA (§4); this order expresses information dependence within the same step and does not mean that the three mechanisms operate only sequentially in real time. For $M_t = (M_{1,t}, M_{2,t}, M_{3,t}) \in \mathcal{M}$, the comprehensive update operator contained in the 7-tuple is

$$\mathrm{Update}(M_t, x_t, e_t) := (M_{1,t+1}, M_{2,t+1}, M_{3,t+1}).$$

Hence the additional arguments appearing in Update_2 and Update_3 are intermediate values generated by earlier component updates within the same step, not independent external inputs to the comprehensive map.

(3) *Reference to the one-step update quantity.* The one-step change in the internal model caused by Update is described by $\delta_{M,c,t}$, $\boldsymbol{\delta}_{M,t}$ and δ_t of def:tipa_component_change, which this subsection does not redefine. $\boldsymbol{\delta}_{M,t}$ is used as the primary description of change and δ_t as a secondary aggregate index of the largest one-step change among the three; δ_t is not a quantity fully expressing the component-wise change pattern of the whole internal model.

The environment argument of Update **(rem:update_env_argument).** The environment argument of Update is the realized external environment state $e_t \in E_{\text{ext}}$, not the internal environment estimate $\hat{e}_t$ (def:internal_env_estimate). This is a Phase 1 abstraction defining Update at the level of the system description. At the mechanism level, whether each mechanism actually refers to the realized state or to the internal estimate is treated in §4. The comprehensive type here does not impose observation-mediated dependence; §5 (subsec:s5_observation) states it explicitly as an additional assumption. Also, the space $\mathcal{R}_{\text{int}}$ of internal representations (def:three_primitives) is not among the independent arguments of Update, for the reason given in rem:M_encodes_representation; the concrete information path from internal representation to internal-model update is treated in §4.

Definition (def:update_consistent_sequence): Update-consistent sequence. A sequence $(M_t, x_t, e_t)_{t\geq 0}$ with $M_t \in \mathcal{M}$, $x_t \in X$, $e_t \in E_{\text{ext}}$ is called an **Update-consistent sequence** when, at every time t, $M_{t+1} = \text{Update}(M_t, x_t, e_t)$. Unless otherwise stated, when this subsection speaks of changes in the internal model and indices concerning updates, the object is an Update-consistent sequence. The definition of Update consistency does not include as a requirement that $\text{WF}_{i,t} = 1$ hold at each time (def:wf_realization).

Learning stop on a sequence

Definition (def:update_stop_on_trajectory): learning stop on a sequence. Along an Update-consistent sequence, put $\delta^{\text{L}}_{c,t} := d^{\text{L}}_c(M^{\text{L}}_{c,t+1}, M^{\text{L}}_{c,t})$ and $\delta^{\text{L}}_t := \max_{c=1,2,3} \delta^{\text{L}}_{c,t}$, the one-step change of the learning component (classification C). When $\delta^{\text{L}}_t = 0$ for all $t \in \mathbb{N}_0$, **learning stop** holds on that sequence; this is equivalent to $M^{\text{L}}_{c,t+1} = M^{\text{L}}_{c,t}$ for all c, t. Changes of the working component are not excluded, and learning stop is not identified with $\delta_t = 0$. We have $0 \leq \delta^{\text{L}}_t \leq \delta_t \leq 1$; invariance of the whole state implies learning stop, but the converse does not hold in general. If, over the whole map, the learning projection is the identity, i.e. $\text{pr}_L \text{Update}(m, x, e) = \text{pr}_L m$ for all $(m, x, e) \in \mathcal{M} \times X \times E_{\text{ext}}$, then learning stop holds on any consistent sequence. The paper does not state, as a general consequence, this map-wide property from learning stop on one sequence. Here $\text{pr}_L : \mathcal{M} \to \mathcal{M}^{\text{L}}$ is the coordinate projection.

Time-average internal-model change

Definition (def:update_quality): time-average internal-model change (classification C). The quantities $\lambda_{T,t}$, $\lambda_{M,c,T,t}$ and $\boldsymbol{\lambda}_{M,T,t}$ below are defined along one fixed Update-consistent sequence for the target ACS Σ. The 7-tuple Σ does not uniquely fix the initial state x_0, the initial internal-model state M_0, the environment realization (def:env_realization) or the action sequence (rem:exogenous_environment); these indices are therefore not determined by Σ alone. The paper omits in notation the dependence on the evaluated sequence; the Σ in parentheses identifies the target ACS and does not mean the values are uniquely determined by the 7-tuple alone.

For start time t and window length T, the **maximum-component time-average update rate** is

$$\lambda_{T,t} = \frac{1}{T} \sum_{s=t}^{t+T-1} \delta_s,$$

where δ_s is the maximum of the TIPA component-wise changes $\boldsymbol{\delta}_{M,s}$ at time s. $\lambda_{T,t}$ is therefore not a total change adding the changes of all three mechanisms, but the time average of the maximum component change at each time. For the same window, the **component-wise time-average update rate** is

$$\lambda_{M,c,T,t} := \frac{1}{T} \sum_{s=t}^{t+T-1} \delta_{M,c,s}, \quad c = 1, 2, 3,$$
$$\boldsymbol{\lambda}_{M,T,t} := (\lambda_{M,1,T,t}, \lambda_{M,2,T,t}, \lambda_{M,3,T,t}).$$

$\boldsymbol{\lambda}_{M,T,t}$ is the primary index describing component-wise change without aggregation, and $\lambda_{T,t}$ the scalar index aggregating the maximum component change; they are used for distinct purposes. They are introduced for analysis and the ACS is not required to hold them. When the

start time is 0, $\lambda_T := \lambda_{T,0}$, $\lambda_{M,c,T} := \lambda_{M,c,T,0}$, $\boldsymbol{\lambda}_{M,T} := \boldsymbol{\lambda}_{M,T,0}$. These are descriptive quantities of the internal-model change that actually occurred in the window, and do not express the maximum update rate or other capacity that the ACS could in principle attain (§2.12). Empirical measurement and estimation for concrete implementations or observed data are Phase 2 and later matters.

Environment-change response function; candidate for a constitutively necessary function (`rem:env_change_response`). By §2.11, the standard map forms of the primitive functions (`def:three_primitives`), the internal model structure M (`def:internal_model`) and the update operator Update (`def:update_operator`) are all in place within §2. The remark records, as the **environment-change response function**, a function that can hold across them: a cross-cutting function by which, in response to changes of the external environment state, the ACS can change its current or future response by using some or several of the available sensing, internal-representation formation, evaluation, prediction, internal-model update and action formation. The remark defines no new map, variable or response-time quantity.

This function is at a different theoretical level from the required environmental factors $\eta \in E$ (§2.6). η describes what must be satisfied, in the pair of internal state and external environment state, for the ACS to persist, and its dependence on the external environment state is constrained by (CR-1c). Whether a given required environment can be utilized on the self side (the required-environment utilization function, `rem:env_utilization_function`) can be expressed as the dependence of η on the internal state. What the present remark describes is, by contrast, how the ACS side can respond using its available functions when the external environment changes. This function is therefore not a fifth label to be added to the standard coarse classes Cls_η of `rem:eta_coarse_classes`, and it is not listed in the same list. The following limitations apply.

- (a) No fixed sequence of processing is required. Passing through every stage from sensing to action at every time is not required; both an acute response, from detection of change to a protective response to a locally acute abnormality, and an adaptive response, which after discrimination and evaluation updates the internal model as needed and changes later responses, can be included. Standard functional forms corresponding to each are in §4.
- (b) The remark does not require, for every $m \in \mathcal{M}$, $x \in X$ and $e, e' \in E_{\mathrm{ext}}$, that $e \neq e' \Rightarrow \mathrm{Update}(m, x, e) \neq \mathrm{Update}(m, x, e')$. There are cases in which no internal-model update is needed even if the external environment state changes, and cases in which a response is realized by changing the current action only. This is consistent with the paper's policy of not generally requiring non-degenerate dependence of f on its action argument (`ass:f_decomposition`, §2.7) and with its treatment not excluding identity updates (`def:update_stop_on_trajectory`).
- (c) From this remark the paper does not state, as a general consequence, maintenance of well-formed realization ($\mathrm{WF}_{i,t} = 1$), feasibility $\Pi(x_t) \geq 0$, non-blockage $A_t \neq \emptyset$ or non-collapse. That is, from the existence of a function able to respond, success of persistence is not stated as a general consequence.
- (d) When a change from outside irrecoverably damages the ACS's structure before a response, that the ACS does not persist is not judged to show the absence of the function able to respond.
- (e) The paper imposes no uniform deadline for responding after an environmental change in Phase 1; the time scale on which room to respond exists depends on the kind of change and the implementation.
- (f) Changes that were not predicted can be included, but since Phase 1 has not formalized a judgment condition and tolerance for "unpredicted" applicable generally to the environment-change response function, "unpredicted change" in this sense is not formally defined (`rem:estimate_gap`). The prediction-consistency error and the two kinds of unpredicted change in §4 are quantities relative to the output and the retained record of M_1 (`def:m1_temporal_features`, `def:prediction_record_check`).

The remark is a record of a candidate for a constitutively necessary function. At this stage

it does none of the following: adding it to Cls_η; adding it to CR-2 (§2.9) as a new class of primitive function; introducing a new constitutive requirement; giving it the classification-A label. Such decisions would require: a formal definition of environmental change relevant to persistence; its distinction from environmental differences requiring no response; a way to exclude changes with no room to respond; treatment of responses involving no internal-model update; an abstract condition common to immediate protective responses and adaptive responses; a condition excluding realizations with identity updates; and a separation of the function able to respond from the success of the response. These are matters for Phase 1.α or Phase 2 (`subsec:phase1a`, within present framework).

2.12 Policy for describing cognitive processing capability

`Reference: subsec:cognitive_capacity_policy`

§2.12 states the description policy adopted in Phase 1 for the cognitive processing capability of an ACS. The paper does not define cognitive processing capability formally in Phase 1, and does not introduce it as a single quantity.

Description policy (`rem:cognitive_capacity_policy`). The paper does not define cognitive processing capability as a single scalar. A formalization would have to distinguish at least three things.

- *Quantitative capacity:* the scale of information or representations that can be retained, integrated and updated per time step.
- *Temporal responsiveness:* the speed with which new information, errors or environmental change are reflected in internal representations and evaluations. In the discrete-time framework, the number of steps (latency) to reach a prescribed integration or correction criterion is a candidate for formalization. Temporal responsiveness is a quantity separate from the scale of processing per time step.
- *Quality of integration:* the degree to which new information and existing internal representations can be integrated and updated while keeping consistency with the state of the object and the objects of evaluation. It can include accuracy of state estimation and evaluation, and detection and correction of errors.

The paper draws the following distinctions. First, capacity — the range attainable under some conditions — is distinguished from realized quantity — change that actually occurred. The time-average internal-model change of `def:update_quality` is a descriptive quantity of the latter and does not express cognitive processing capability. Second, capability is distinguished from the bias in its use. The degree of bias appearing in carrying out the responsibilities of the primitive functions (including the allocation or use of resources, evaluation, attention and so on) is described in §5 as bias (`def:bias_tendency`) and expressed by the bias strength $\beta_{\iota,t}$ (`def:beta_accumulated`), and is not included in cognitive processing capability. Third, no general formula aggregating the three categories into one real number, and no total order among the categories, is placed. Fourth, cognitive processing capability is not identified with the degree of operation expressed by the functional axes of §4, or with the state of the base expressed by the health indices of §5. The remark introduces no new symbol, map, constitutive requirement or assumption. Formalization and measurement of the three categories are Phase 1.α or Phase 2 matters (L3 of §9, `subsec:phase1a`).

2.13 Organization of well-formedness, constitutive requirements and assumptions

§2.13 lists the conventions, constitutive requirements and assumptions introduced in §2. It is not a place where they are newly defined: the formal definition of each lies in the subsection that semantically owns it, and this subsection restates only location and status.

The paper distinguishes three kinds of conditions supporting the definition of an ACS. First, **conventions**: agreements the paper adopts for description, not conditions the ACS must

satisfy. Second, **constitutive requirements**: semantic conditions that the realization of the 7-tuple structure must satisfy to be recognized as an ACS in this paper. Third, **assumptions**: conditions introduced to limit the range in which later propositions hold. Conventions and constitutive requirements are not included in the list of assumptions.

Conventions. Two are restated as independent items. The **convention on levels of description** (§2.4, `subsec:acs_7tuple`): it distinguishes the abstract structure type, its realization as an individual Σ_i, and time-indexed values, and writes a quantity depending on an individual's realization as a time-independent map plus its time-indexed value, in two steps. The **type separation convention TSC** (§2.2, `subsec:wellformedness`): a convention adopted to separate the internal-state type X and the external-environment-state type E_{ext} also in their set representations; a tagged, disjoint representation is a means of implementing it, and what TSC requires is the type separation itself.

Constitutive requirements. Two. **CR-1** (required-environment consistency, §2.6): each element of E is truly necessary for the ACS's persistence (CR-1a); the whole of the required factors can be satisfied simultaneously (CR-1b); and, for each factor, with the internal state at the starting time fixed, lasting non-satisfaction can be realized in which satisfaction and non-satisfaction are separated by differences of external environment state (CR-1c). **CR-2** (possession of primitive and base functions, §2.9): basic functions are possessed for each of sensing, discrimination, evaluation and action (CR-2a); they are functionally non-degenerate (CR-2b); and the base functions supporting their activity are possessed (CR-2c). CR-1 and CR-2 are both semantic conditions on a realization, not conditions mechanically checkable from the 7-tuple. Their objects differ, and the paper does not state either as a general consequence of the other. The paper gives no procedure to judge their satisfaction from finite observation in Phase 1.

List of assumptions. Nine assumptions are introduced in §2. The status symbols are: A for axiomatic, S for structural, O for observational. The content, status and strength of each assumption are as stated in the subsection where it is located; "main dependents" shows the main definitions, propositions and remarks referring to the assumption, not all references. The manuscript gives these in a table; the entries are as follows.

Assumption	Location	Main dependents	Status	Strength
`ass:no_branching`	§2.5 (structural self and temporal self-identity)	`def:time_identity`	A	strong
`ass:compact_spaces`	§2.7 (state transition)	`def:acs`, `prop:At_compact`, `def:x_E`, `def:update_section5`	S	strong
`ass:f_continuity`	same	`def:acs`, `prop:At_compact`	S	medium
`ass:Eext_time_fixed`	same	`rem:fixed_environment_and_state_dependent_viability`	S	medium
`ass:f_decomposition`	same	`def:acs`, `rem:oplus_neutrality`, `rem:blockage_auto`	S	strong
`ass:env_input_nondegeneracy`	same	`prop:nondegeneracy_implication`	S	weak
`ass:env_viable_nondegeneracy`	§2.8 (feasibility)	`prop:nondegeneracy_implication`, `prop:feasibility_sensitivity_scope`	S	weak
`ass:viable_realizability`	same	`rem:viable_realizability_consequences`	S	medium
`ass:M_compact`	§2.10 (internal model)	`def:model_parameter_space`, `def:update_operator`	S	strong

Those with strength "strong" are four: `ass:no_branching`, `ass:compact_spaces`, `ass:f_decomposition` and `ass:M_compact`. They are assumptions whose relaxation could change the range in which the main definitions of §2 or the main propositions of later sections hold; examination of their weakening or alternative formulations is in §9. The strength "strong" of `ass:M_compact` applies to the nonemptiness and compactness of each $\mathcal{M}_c$; the normalization

of distance scale contained in that assumption is treated as a metric convention and is not included in this strength evaluation.

What §2 does not claim, and delegation to later Phases. §2 defines blockage, functional termination and collapse, but by itself claims none of the following: reaching blockage, functional termination or collapse; final loss of well-formed realization; final loss of the structural self; social interaction; stochastic stability; implementability; measurability. The following are not formalized within Phase 1; each is delegated to a later Phase or reserved as a candidate for future extension or reconsideration, and the paper does not claim that all of them will be formalized in later Phases.

1. Stochastic extension, continuous time, and generalization to non-compact internal state spaces.
2. Empirical identification procedures for required environmental factors and the treatment of identification error (`rem:necessity_identification`); existence of necessity and identifiability are distinguished.
3. Reformulation of temporal self-identity when branching is allowed (relaxation of `ass:no_branching`, §9).
4. An operational procedure to judge the value of the well-formed realization indicator $\mathrm{WF}_{i,t}$ (`def:wf_realization`) from finite observation. The relation of WF to the functional-termination indicator $\mathrm{FT}_{i,t}$ and the collapse indicator $\mathrm{Col}_{i,t}$ was stated in `rem:collapse_wf_relation`.
5. The general form of the judgment rules for conditions (FT-1), (FT-2) of functional termination and (SC-1), (SC-2) of collapse, and the general form of the organizational relations that establish identity as a structural individual referred to by (SC-1) (`def:functional_termination`, `def:collapse`). The paper treats these as fixed per target system.
6. Dynamic reorganization of the self–environment boundary, and extension to a time-dependent external environment state space.
7. The magnitude of the action difference (`def:action_difference`), multi-step action, comparison of the contributions of actions and of the action-response component, and the balance of resource quantities common to outside and inside. The generation rule of the background sequence and coupled dynamics with other ACS individuals are Phase 2 matters (`rem:exogenous_environment`; L2 of §9).
8. Quantification of the gap between the realized external environment state e_t and the internal environment estimate $\hat{e}_t$ (`rem:estimate_gap`).
9. Where needed, introduction of an explicit encoding map from the space $\mathcal{R}_{\mathrm{int}}$ of internal representations to the internal-model state space $\mathcal{M}$ (`rem:M_encodes_representation`); the paper does not promise to introduce such a map in later Phases.
10. Formalizing the cross-cutting responsiveness to environmental change (`rem:env_change_response`) as a new constitutive requirement. Persistence relevance of environmental change, the time scale on which room to respond exists, judgment of unpredicted change, routes of repair, substitution and recovery, and the necessary and sufficient conditions for raising that capability to a condition for being an ACS, are objects of study in Phase 1.α or Phase 2 (`subsec:phase1a`).
11. Fixing the granularity of classification of required environmental factors (`rem:eta_coarse_classes`). The standard coarse classes Cls_η are not a partition, and their exhaustiveness and exclusiveness are not proved.

Connection to the three principles. §2 defines an ACS as the 7-tuple $\Sigma = (X, U, E, f, \Pi, M, \mathrm{Update})$ and gives the framework of description relative to the external environment state space E_{ext}, the structural self, well-formed realization and collapse, the required environment space, state transition, persistence evaluation and feasibility, primitive functions, the internal model structure, internal-model update, and the policy for describing cognitive processing capability. §3 formally defines, under these definitions, the three principles — self-maintenance (SM), environmental utilization (EU) and feasible response (FR) — distinguishing the conceptual definition of each principle from its time-indexed satisfaction predicate, and treats conditional implications and non-implications among the time-indexed satisfaction predicates.